Avyukth Harish

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Education

University of California, Berkeley Expected Graduation May 2028

B.A. in Computer Science, Data Science, Mathematics (Dean's List)

Relevant Coursework: Machine Learning, Numerical Analysis, Multivariable Calculus, Analytic/Quantum Mechanics, Stochastic Control, PhD Statistics, Real Analysis, Data Structures and Algorithms, Linear Algebra, Probability, Graduate NLP

Course Staff: Probability & Random Processes, Introduction to Machine Learning, Discrete Math & Probability

Skills

Programming: Python, Java, C, Pandas, PyTorch, TensorFlow, CUDA, SciPy, gpCAM, Triton, Git, Hugging Face, MuJoCo

Research & Projects

Research Affiliate | International Computer Science Institute July 2026 – Present

- Authoring a survey of physical-time conditioning in diffusion/flow-matching transformers (AdaLN/Fourier, lag-sampling, time-coupled diffusion, attention embeddings) for weather, molecular dynamics, and PDE forecasting (paper in preparation).
- Unifying the formulations of GEM-3, Implicit Transfer Operators, and dynamics-informed diffusion. Designing an attention-based time-conditioning mechanism for DiTs and an ablation suite on simulated Lorenz '96 data.

Research Affiliate | Lawrence Berkeley National Lab January 2026 – Present

- Formulated non-stationary GP kernels (Gibbs/Paciorek–Schervish, deep kernel) that learn location-dependent variance and length-scales from data, replacing stationary Matérn kernels on spatially heterogeneous 1D and 2D scientific fields.
- Achieved up to 30% lower RMSE than stationary baselines across chirp, topography, and X-ray scattering datasets; identified stationary-kernel failure modes that miscalibrate uncertainty and mislead acquisition in Bayesian experimental design.

Research Assistant | Berkeley AI Research Lab January 2025 – Present

- Examined the negative effect of weight tying in 5+ LLMs through geometric subspace and gradient scaling experiments.
- Collaborating with Meta FAIR to build a brain-routed Mixture-of-Blocks LM, routing Qwen3-0.6B blocks via sparsemax over 8 Qwen3-8B-derived fMRI voxel clusters; an LM-only router recovered brain routing at 4.4× chance (paper in preparation).
- Restored specialization lost in SFT with a custom mutual-information routing loss, raising I(token; block) from 0.2 to 1.38.

Systems & Inference Optimization for Transformer Serving June 2026 – August 2026

- Built nanoGPT benchmarks for profiling latency and memory footprint across naive, KV-cached, and custom-kernel paths.
- Implemented a custom, verified Triton FlashAttention kernel using block-tiled computation and online softmax.
- Extended infrastructure with continuous batching and paged KV-cache memory management for concurrent request handling.

Latent Dynamics in Diffusion Models December 2025 – January 2026

- Engineered VAE compressing MNIST to 16D latent space and trained diffusion model collecting 100+ denoising trajectories.
- Applied SINDy (sparse regression) with polynomial/exponential libraries discovering interpretable equation dynamics.

Work Experience

Data Engineer Intern | Coinbase May 2026 – August 2026

- Boosted asset recommendation ranking quality by implementing a custom contrastive InfoNCE objective with selected negative samples over user-asset embeddings, improving MRR by 50% over a binary cross-entropy baseline.
- Built agentic automation of column-level security workflows and embedding-based classifier for PII in Snowflake databases.
- Reduced warehouse clutter by flagging 50+ redundant tables via a deterministic detection engine and a headless Claude agent.

Software Engineer (Contract) | Google DeepMind February 2026 – May 2026

- Developed an agentic framework and graph package to automate I/O stitching between state-of-the-art bioML models.
- Created a PyPI package and tool registry for bioinformatics software engineering pipelines, and automated 10+ pipelines.

AI/ML Intern | Arista Networks February 2025 – December 2025

- Developed GCP document processing pipeline using 7000+ PDFs and utilized Vertex AI hybrid search for semantic retrieval.
- Retrieved relevant chunks using RAG and generated LLM responses with hallucination detection for source-backed answers.
- Curated structured dataset with 10,000+ document segments and fine-tuned Vertex AI baseline model for enhanced accuracy.

Publications

Weight Tying Biases Token Embeddings Towards the Unembedding Space (arXiv)
A. Lopardo, **A. Harish**, C. Arnett, A. Gupta - ACL Findings 2026

Cross-lingual Fairness Drift In LLM Moral Reasoning
E. Xie, A. Chang-Lee, **A. Harish**, A. Vaidheeswaran - ICLR CAO Workshop 2026

The Critical Impact of Non-Stationarity on Bayesian Decision Making for Self-Driving Labs
M. Noack, A. Gongora, **A. Harish** (In progress)